ALzheimer’s Disease

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Why should we care about Alzheimer’s Disease?

AD is a complex, progressive neurological disease associated with **cognitive, functional, and behavioural** changes

Globally, an estimated **43.8 million people** are living with AD and other dementias costing ~1.3 trillion dollars per year

AD can be a **debilitating disease**, resulting in memory loss and severe interference with activities of daily living

There is **only one disease-modifying therapy for AD** at this time; currently, other treatments are symptomatic

AD often makes activities of daily living (ADLs) quite difficult for patients

- Although subtle losses of memory are observed in very early AD, as the disease progresses, symptoms begin to impact a person's ability to function as an independent adult

Impact on patients:
Earlier stages of AD

- Planning dinners
- Handling finances and paperwork
- Shopping
- House-keeping
- Remembering appointments
- Transportation

Memory-based tasks are impacted early

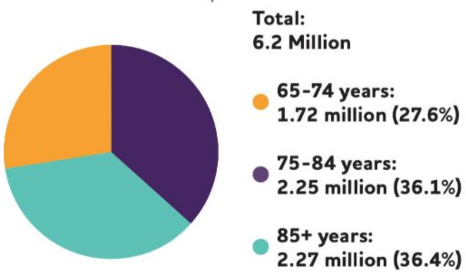
Impact on patients:
Later stages of AD

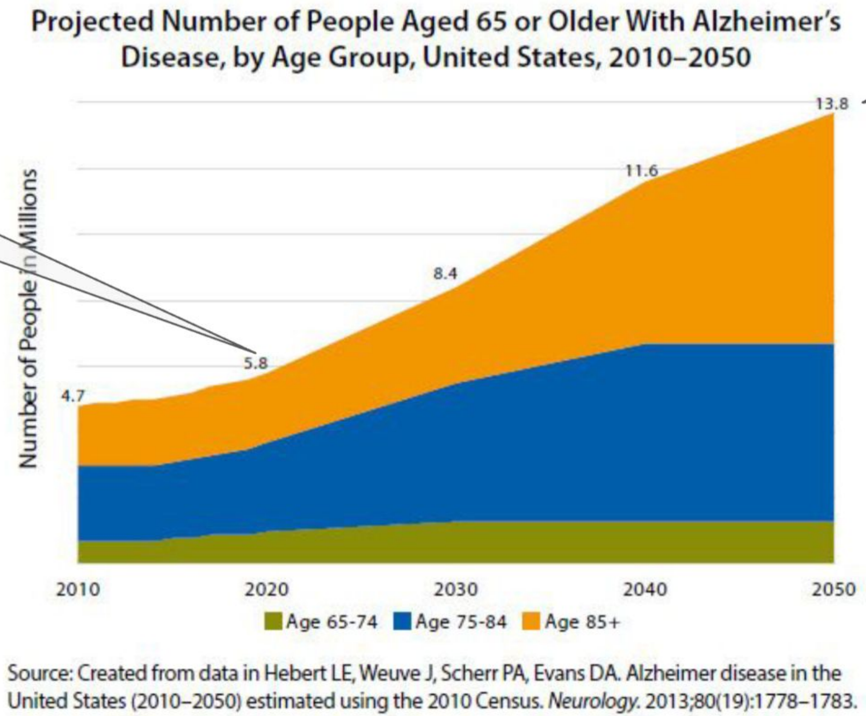
- Bathing
- Dressing
- Toileting
- Feeding
- Walking
- Mood swings

All basic activities and personality traits are impacted later on

It impacts a lot of people: prevalence and epidemiology of AD

Current 2021 prevalence: 6.2 million US AD patients





Projected 2050 prevalence: ~14 million US AD patients

Accounts for 60-70% of all dementia cases: still more people to care for

[1] Alzheimer's Association. Accessed August 24, 2021 from Website

How deep learning and artificial intelligence can help AD

Artificial Intelligence (AI), especially using technologies like convolutional neural networks (CNNs), has significant potential in enhancing clinical approaches to Alzheimer's disease (AD). Here are some ways AI can be used:

Early Detection and Diagnosis:

- AI analyzes MRI data to detect early signs of Alzheimer's, enabling timely intervention.

Monitoring Disease Progression:

- AI tracks changes in brain structures like the hippocampus over time, offering precise assessments of disease progression.

Personalized Treatment Plans:

- AI predicts patient responses to treatments based on genetic markers and disease patterns, supporting personalized care.

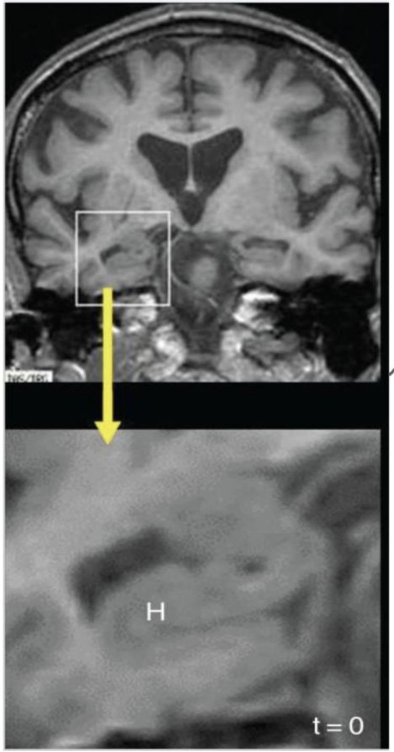
Improving Clinical Trials:

- AI identifies optimal candidates for clinical trials, enhancing research efficiency and effectiveness.

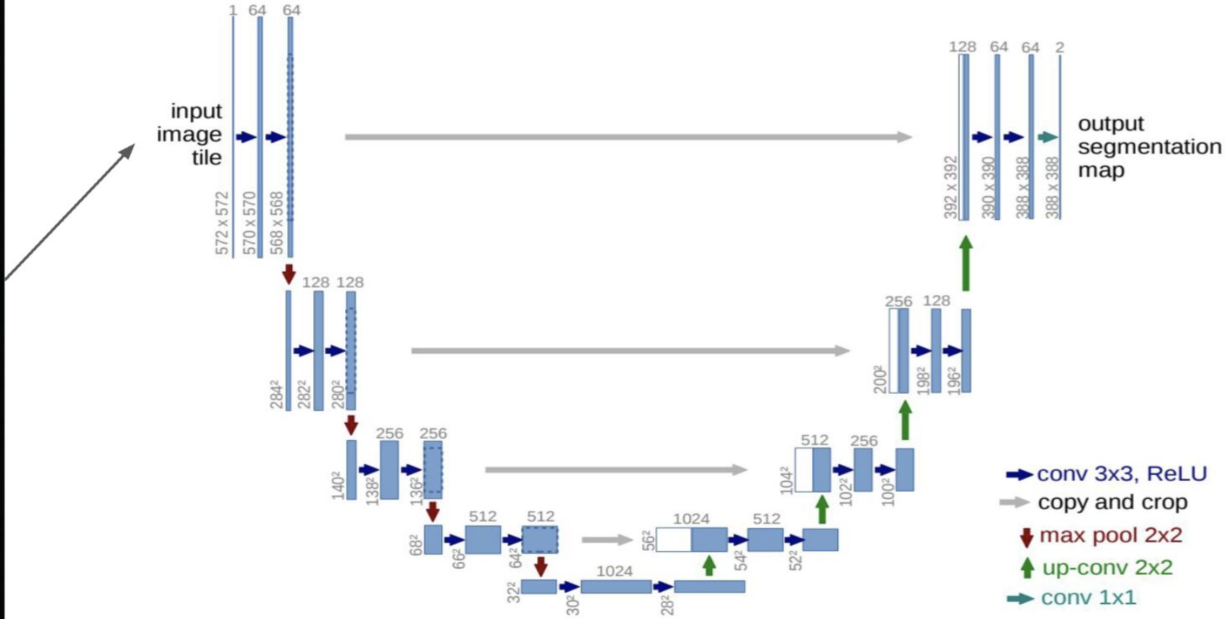
Enhancing Cognitive Assessments:

- AI-driven tools provide consistent, precise measurements of cognitive function, aiding in monitoring and treatment adjustments.

How could AI potential be used in the clinic for AD ?



Convolutional neural network



Using MRI

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